

PocketSmart AI

Project Development Phase

1. Development Overview

PocketSmart AI was developed as a web-based GenAI budget and recommendation assistant using Python and FastAPI.

2. Backend Development

The backend was developed using FastAPI. It handles user authentication, planner requests, input validation and recommendation generation.

3. Frontend Development

The frontend was developed using HTML, CSS, JavaScript and Jinja2 templates. It provides the homepage, planner forms, login/register pages and recommendation result pages.

4. Database Development

SQLite is used to store user information and application data.

5. AI Integration

Google Gemini API is integrated to generate personalized recommendations based on user requirements, preferences and budget.

6. Recommendation Modules

The system contains three main planning modules:

- Home Decor Planner
- Party Planner
- Jewelry Planner

7. Fallback System

A fallback recommendation system is included so that the application can provide predefined recommendations when the AI service is unavailable.

8. Authentication

The application provides:

- User registration
- User login
- User logout
- Session-related functionality

9. API Endpoints

Main planner endpoints include:

- `/planner/home`
- `/planner/party`
- `/planner/jewelry`

Authentication endpoints include:

- `/auth/register`
- `/auth/login`
- `/auth/logout`
- `/auth/session-info`

10. Development Tools

- Python
- FastAPI
- HTML
- CSS
- JavaScript
- Jinja2
- SQLite
- Google Gemini API
- Visual Studio Code
- GitHub

11. Development Outcome

The developed system allows users to enter their budget and preferences and receive suitable recommendations for home decor, party planning and jewelry through a simple web interface.